

Rudranarayan

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Mangalore, Karnataka, India

Summary

Aspiring AI Research Engineer with a strong foundation in autonomous systems, multi-agent reinforcement learning, and evolutionary computation. Experienced in designing novel AI architectures — from emergent economic agents to self-evolving neural networks. Passionate about pushing the boundaries of artificial intelligence through research-oriented, self-driven projects and real-world internship exposure.

Education

NITTE Institute of Professional Education

Bachelor of Computer Applications — Artificial Intelligence (BCA-AI)

Mangalore, India

2023 – 2027

Experience

AI/ML Intern

May 2025 – July 2025

CSD, National Institute of Technology Karnataka (NITK Surathkal)

- Developed CNN, ANN, and NEAT models through research-oriented virtual AI laboratory exercises.
- Benchmarked and compared deep learning architectures using Python, TensorFlow, and PyTorch.
- Analyzed model performance across classification tasks and contributed to evaluation pipelines.

Web Developer (Contract)

2025

Vidman Homey Huts

Remote

- Designed and deployed a full production website with Next.js for a homestay business client.
- Implemented secure backend authentication and environment variable management.

Projects

ECHO-5: AI 5v5 Tactical Multi-Agent Arena | *Python, Reinforcement Learning, Multi-Agent Systems*

- Engineered a full 5v5 multi-agent combat simulation where AI agents learn tactical team coordination autonomously.
- Implemented RL with reward shaping, enabling emergent cooperative and adversarial strategic behaviors.

AetherMarket: AI Economic Simulation | *Python, Agent-Based Modeling, Simulation*

- Built an AI-driven market simulation where autonomous economic agents negotiate supply, demand, and pricing.
- Modeled complex emergent market dynamics through multi-agent interaction and adaptive decision-making.

Self-Driving AI Cars | *Python, NEAT, Evolutionary Neural Networks*

- Simulated autonomous vehicle navigation using NEAT — neural networks evolved through genetic selection.
- Cars iteratively learned obstacle avoidance and lane navigation across generations without hard-coded rules.

Evolving Double Pendulum | *Python, Physics Simulation, Neural Evolution*

- Modeled chaotic double-pendulum dynamics and applied evolutionary algorithms to predict motion trajectories.
- Demonstrated neural network convergence on deterministic chaos — a novel research-oriented experiment.

Technical Skills

Languages: Python, Java, C

AI/ML Frameworks: TensorFlow, PyTorch, Scikit-learn, OpenCV, MediaPipe, LangChain

Research Areas: Reinforcement Learning, Multi-Agent Systems, Evolutionary Algorithms, NEAT, Neural Architecture Search, LLMs, RAG

Tools: Git, GitHub, Docker, n8n, Jupyter Notebook

Achievements

- **Aethron 2K25** — *1st Place*, Hackathon (2025)